ZOO 302 Current Final Term Spring 2021

Q1. WHAT IS OOGENESIS? 2 Marks

Oogenesis is the formation of ovum or egg. In human oogenesis starts in the ovaries of the fetus before birth.

Q2. Define Parthenogenesis? 2 Marks

Parthenogenesis is a form of asexual reproduction in which development of egg occur without being fertilized.

Q3. Placenta. 2 Marks

The placenta is the organ that sustains the embryo and fetus throughout the pregnancy and through which gases are exchanged, nutrients are supplied and wastes are removed from the fetal systems to the maternal end.

Q4. Function of Mesonephros? 3 Marks

It is a paired organ consisting of a set of nephrons having capsules that filter blood from the glomerulus and tubules whose cells reabsorb water and nutrients and secrete nitrogenous wastes. It serves as the main excretory organ of aquatic vertebrates and as a temporary kidney in reptiles, birds, and mammals.

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Q5. Explain the structure of Sperm? 3 Marks

A mature human sperm consists of a head, midpiece, and tail. The head is flat, disc shaped 5.1 µm by 3.1 µm and contains the haploid nucleus, which is mostly DNA. The acrosome, a cap over most of the head, contains an enzyme called acrosin that assists the sperm in penetrating the

outer layer surrounding a secondary oocyte. The spiral mitochondria in the midpiece supply the ATP necessary for these movements. The sperm tail is 50 µm long contains an array of microtubules that bend to produce whip like movements.

Q6. Osmoregulation and its function? 5 Marks

A variety of mechanisms have evolved in vertebrates to cope with their osmoregulatory problems, and most of them are adaptations of the urinary system. Vertebrates have a closed circulatory system containing blood that is under pressure.

This pressure forces blood through a membrane filter in a kidney, where the following three key functions take place:

• 1. Filtration

• 2. Reabsorption

• 3. Secretion

1. Filtration: In this process blood passes through a filter that retains blood cells, proteins, and other large solutes but let's small molecules, ions, and urea pass through.

2. Reabsorption: In reabsorption selective ions and molecules are taken back into the bloodstream from the filtrate.

3. Secretion: In this process select ions and end products of metabolism (e.g., K, H, NH3) that are in the blood are added to the filtrate for removal from the body.

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Q7. Methods of Sexual Reproduction? 2 Marks

The various forms of sexual reproduction are:

1. Hermaphroditism

2. Sequential hermaphroditism

Q8. What is Pronephros? 3 Marks

The Pronephros:

• Appears only briefly in many vertebrate embryos, and not at all in mammalian embryos.

• In some vertebrates, the pronephros is the first osmoregulatory and excretory organ of the embryo as in tadpoles and other amphibian larvae.

In vertebrates such as in hagfishes, pronephros remains as the functioning kidney.

• The primitive pronephric kidney is found in adult hagfishes and in the embryos of fishes and amphibians.

Q9. Types of Parthenogenesis? 5 Marks

Types of Parthenogenesis:

• Diploid Parthenogenesis

• Haploid Parthenogenesis

Diploid Parthenogenesis:

• In this type, no meiosis occurs and the eggs are formed by mitosis.

• The eggs develop and the offspring are the copies or clones of the parents. In some cases, eggs are formed by modified form of meiosis involving total non-disjunction of the chromosomes thus retaining the diploid number.

• Such eggs develop into young females.

• Examples: some species of flat worm, rotifers and insects.

Haploid Parthenogenesis:

• In this type of Parthenogenesis haploid ova are formed by meiosis and they may or may not be fertilized by sperm.

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• If fertilized, the zygotes develop into females (Queen or sterile female workers in bee).

• If unfertilized the eggs develop to become haploid males (drones). On maturity they produce sperms by mitosis.

Q10. What is fertilization? 2 Marks

Fertilization is the fusion of haploid gametes, egg and sperm, to form the diploid zygote.

Q11. Structure of Testis? 5 Marks

Each testis is covered by a thick capsule called Tunica Albuginea. It is composed mainly of collagenous fibers and smooth muscle cells. Posteriorly the tunica albuginea is thickened and is called mediastinum testis. Thin fibrous septa, called Septula testis, radiate from the midiastinum to the capsule and divide the interior of the testis into 200 to 300 Pyramidal Lobules. The septula are not complete partitions; they show numerous gaps, through which the testicular lobules communicate quite freely. Beneath the tunica albuginea there is a layer of loose areolar connective tissue containing networks of blood vessels. This layer is known as tunica vasculosa, also lines the Septula Testis. Each testicular lobule contains 1-4 highly convoluted Seminiferous Tubules. Each seminiferous tubule is intricately coiled, averaging about 0.2 mm diameter and 50 cm in length. Individual tubules usually commence as free blind ends but neighboring tubules may form anastomosing loops.

Q12. Types of vertebrate kidneys with examples? 5 Marks

Overall, there are three kinds of vertebrate kidneys:

• Pronephros,

• Mesonephros, and

• Metanephros.

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The Pronephros:

• Appears only briefly in many vertebrate embryos, and not at all in mammalian embryos.

• In some vertebrates, the pronephros is the first osmoregulatory and excretory organ of the embryo as in tadpoles and other amphibian larvae.

In vertebrates such as in hagfishes, pronephros remains as the functioning kidney.

• The primitive pronephric kidney is found in adult hagfishes and in the embryos of fishes and amphibians.

Mesonephros

It is a paired organ consisting of a set of nephrons having capsules that filter blood from the glomerulus and tubules whose cells reabsorb water and nutrients and secrete nitrogenous wastes. It serves as the main excretory organ of aquatic vertebrates and as a temporary kidney in reptiles, birds, and mammals. The mesonephros is the functional embryonic kidney of many vertebrates and also adult fishes and amphibians. It is structurally similar to the non-segmented opisthonephric (advanced mesonephric) kidney of most nonamniote vertebrates, such as sharks.

Metanephros

Member of the final and most caudal pair of vertebrate renal organs that functions as a permanent adult kidney is in reptiles, birds, and mammals. Not present at all in lower forms. The physiological differences between these kidney types are primarily related to the number of blood-filtering units. The pronephric kidney is formed in the anterior portion of the body cavity and contains fewer blood-filtering units than either the mesonephric or metanephric kidneys.

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Q13. Methods of Asexual Reproduction? 2 Marks

• Following are the forms:

• Fission

a. binary fission

b. multiple fission

• Budding (both internal and external)

• Fragmentation

• Parthenogenesis

Q14. Structure of Ovary? 3 Marks

The ovaries are held in place by ligaments (mesovarium) attached to the uterus through which vessels and nerves pass to the ovary.

The ovaries have two functions:

• 1. Production of ova

• 2. Secretion of hormones.

The surface of the ovary is covered by a single layer of cuboidal epithelium, called germinal epithelium. Fibrous connective tissue forms a thin capsule, the tunica albuginea, immediately beneath the epithelium. Ovary is divided into an outer cortex and an inner medulla. The cortex consists of a very cellular connective tissue stroma in which the ovarian follicles are embedded. The medulla is composed of loose connective tissue, which contains blood vessels and nerves.

Q15. Process of Fertilization? 5 Marks

• Fertilization involves four events:

• i) the sperm contacts the egg and recognition occurs.

• ii) the sperm or sperm nucleus enters the egg.

• (iii) the egg becomes activated, and certain developmental changes begin.

• (iv) the sperm and egg nuclei fuse.

An egg is surrounded by a plasma membrane and by one or more external coverings that are important in fertilization. For example, a mammalian egg is enclosed by a thick, noncellular,

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zona pellucida, which is surrounded by a layer of granulosa cells derived from the follicle in which the egg develops. Before fertilization the mammalian sperm undergoes capacitation, a maturation process in the in the female reproductive tract. During capacitation, the sperms become increasingly motile and capable of undergoing an acrosome reaction when they encounter an egg.

Sperm Entry

The plasma membrane of the egg is covered with microvilli, which elongate to surround the head of the sperm. As they do so, the plasma membranes of sperm and egg fuse and a fertilization cone is formed that contracts to draw the sperm into the egg.

Fertilization activates the egg

Release of calcium ions into the egg cytoplasm triggers the activation program, a series of metabolic changes within the egg. Aerobic respiration increases, this causes a burst of protein synthesis after sperm entry.

Sperm and egg pronuclei fuse, restoring the diploid state.

After the sperm nucleus enters the egg, it swells and forms the male pronucleus. The nucleus formed during completion of meiosis in the egg becomes the female pronucleus.

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Q16. Stages of Development before Implantation? 5 Marks

First Trimester

After fertilization, usually in the upper third of the uterine tube, the zygote goes through several cleavages as it moves down the tube. It eventually becomes a solid ball of cells called a morula, and by the fourth day, it develops into a 50 to120-cell blastula stage called a blastocyst. The next stage of development occurs when the blastocyst adheres to the uterine wall and implants. During implantation, the outer cells of the blastocyst, called the trophoblast, invade the endometrium. Only the inner cell mass gives rise to the embryonic body. Eventually, these cells arrange in a flat sheet that undergoes a gastrulation similar to that of reptiles and birds. Once gastrulation is completed, the rest of the first trimester is devoted to organogenesis and growth.

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